ICT4153

Mobile Application Development

Practical 01

Introduction to Flutter

- **What is Flutter?**

Flutter is an open source framework developed by Google to create high quality, high performance mobile applications across mobile operating systems - Android and iOS. It provides a simple, powerful, efficient and easy to understand SDK to write mobile applications in Google's own language, Dart.

With Flutter, we can write the app's code once and deploy it on multiple platforms(Android, IOS, Desktop). Due to its flexibility and performance capabilities, Flutter is the top pick for modern app development across various platforms.

In general, developing a mobile application is a complex and challenging task. There are many frameworks available to develop a mobile application. Android provides a native framework based on Java language and iOS provides a native framework based on Objective-C / Swift language.

However, to develop an application supporting both the OSs, we need to code in two different languages using two different frameworks. To help overcome this complexity, there exists mobile frameworks supporting both OS. These frameworks range from simple HTML based hybrid mobile application framework (which uses HTML for User Interface and JavaScript for application logic) to complex language specific framework (which do the heavy lifting of converting code to native code). Irrespective of their simplicity or complexity, these frameworks always have many disadvantages, one of the main drawback being their slow performance.

- **What Makes Flutter Unique?**

Flutter has several features and design principles that set it apart from other frameworks. It neither uses WebView nor the OEM widgets; instead, it uses customizable widgets, which enhances the application's interface. Also, the hot reload feature lets developers make changes in the app without restarting.

- **Applications Made with Flutter Framework**

There are quite a few popular applications that are made with the Flutter framework. Few of the are −

- Google Products - Google Ads, Google pay
- Reflectly - Personal Journal App
- Alibaba's Xianyu - Ecommerce Platform
- eBay Motor
- Nubank
- Hamilton

➢ Mobile App Development with Flutter

Flutter is an open-source UI toolkit developed by Google that is used to build, network, and integrate apps from a single code base. Following are the steps that would take you through the process of developing, testing and deploying a mobile application −

- Set Up the Development Environment
- Create a New Flutter Project
- Build your app's interface by defining Widgets
- Implement application logic
- Add functionality and features
- Test your application
- Debug and Optimize
- Build and Deploy

➢ Features of Flutter

Flutter framework offers the following features to developers −

- Modern and reactive framework.
- Uses Dart programming language and it is very easy to learn.
- Fast development.
- Beautiful and fluid user interfaces.

- Huge widget catalog.
- Runs same UI for multiple platforms.
- High performance application.

➢ **Advantages of Flutter**

Flutter comes with beautiful and customizable widgets for high performance and outstanding mobile application. It fulfills all the custom needs and requirements. Besides these, Flutter offers many more advantages as mentioned below −

- Dart has a large repository of software packages which lets you to extend the capabilities of your application.
- Developers need to write just a single code base for both applications (both Android and iOS platforms). *Flutter* may to be extended to other platform as well in the future.
- Flutter needs lesser testing. Because of its single code base, it is sufficient if we write automated tests once for both the platforms.
- Flutter's simplicity makes it a good candidate for fast development. Its customization capability and extendibility makes it even more powerful.
- With Flutter, developers has full control over the widgets and its layout.
- Flutter offers great developer tools, with amazing hot reload.

➢ **Disadvantages of Flutter**

We have seen earlier that the Flutter has many advantages, but it also contains some disadvantages, which are given below.

- The Flutter is a comparatively new language that needs continuous integration support through the maintenance of scripts.
- It provides very limited access to SDK libraries. It means a developer does not have a lot of functionalities to create a mobile application. Such types of functionalities need to be developed by the Flutter developer themselves.
- The Flutter apps do not support the browser. It only supports Android and iOS platforms.
- It uses Dart programming for coding, so a developer needs to learn new technologies. However, it is easy to learn for developers.

Flutter Application Architecture

Flutter architecture application mainly consists of:

- **Widgets**
- **Gestures**
- **Concept of State**
- **Layers**

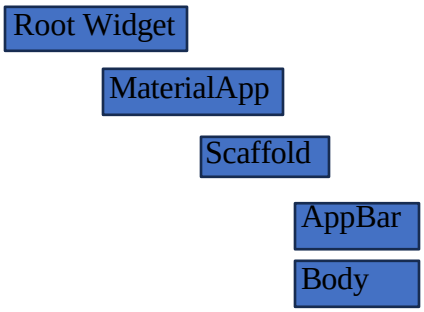
➢ **Widgets**

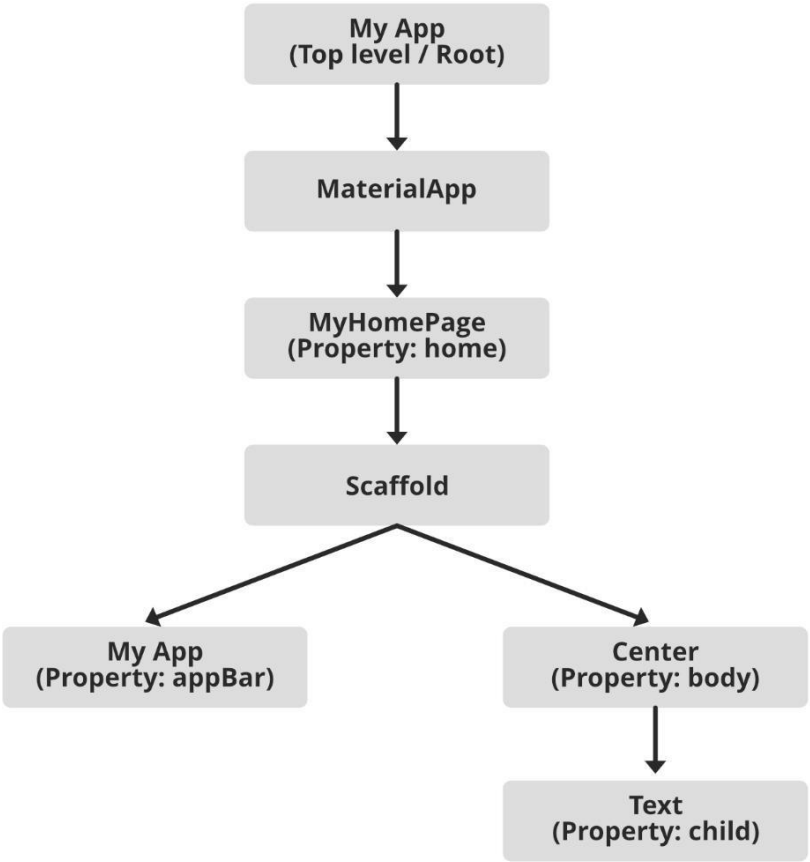
- Widgets are the primary component of any Flutter application, acting as the UI for user interaction.
- Every Flutter application is itself a widget made up of a combination of widgets.
- The root widget defines the structure of the application.
- The MaterialApp widget is responsible for holding the internal components of the application in place.
    - It is where the properties of the UI and the application itself are set.
- The Scaffold widget is a part of the MaterialApp and consists of the visible components (widgets) of the application.
- The Scaffold widget has two primary properties:

✓ Body: Contains the main content and child widgets of the application.

✓ AppBar: Represents the top app bar, typically used for navigation and titles.

- The hierarchy of a Flutter application follows this structure:





- The Scaffold widget usually contains an AppBar widget.
  - The AppBar widget defines the app bar of the application.
- The body of the Scaffold holds all the component widgets of the application.
  - This is where the properties of the widgets are set.
- These widgets collectively form the Homepage of the application.
- The Center widget is often used within the body and has a child property.
  - The child property refers to the actual content.
  - The content is commonly created using the Text widget.

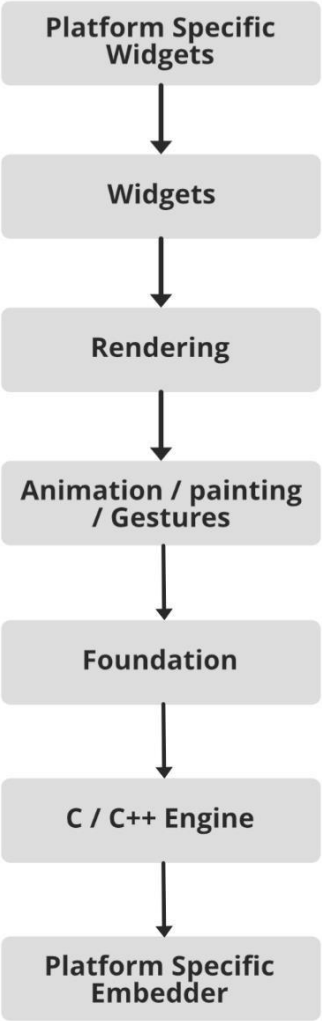
➢ **Layers**

- The **Flutter framework** is categorized into **layers** based on complexity, forming a hierarchical structure.
- **Layers are built on top of each other**:

1. **Topmost layer**:

    - Contains widgets specific to the operating system (e.g., Android or iOS).
2. **Second layer**:
    - Consists of **native Flutter widgets**:
        - Structural UI components.
        - Gesture detectors.
        - State management components.
3. **Third layer**:
    - Handles **UI and state rendering**.
    - Includes all the visible components of a Flutter application.
4. **Next layer**:
    - Focuses on **animations**:
        - Used in transitions, image flow, and gestures.
5. **Lower layers**:
    - Deal with **high-level system design**, which is not the focus here.

Each layer decreases in complexity as it moves down the hierarchy.



- **Gestures**

Gestures are the primary means of physical interaction with a Flutter application.

GestureDetectors are used to handle gestures:

- ✓ They are **invisible widgets** that process physical interactions.

Common gestures include:

- ✓ **Tapping**
- ✓ **Dragging**
- ✓ **Swiping**

Gestures can be used to **enhance user experience** by performing desired actions based on simple interactions.

- **Concept of State**

If you have ever worked with React-js, you might be familiar with the concept of a state. The states are nothing but data objects. Flutter also operates on similar turf. For the management of state in a Flutter application, Stateful-Widget is used. Similar to the concept of state in React-js, the re-rendering of widgets specific to the state occurs whenever the state changes. This also avoids the re-rendering of the entire application, every time the state of a widget changes.

The architecture of a Flutter app or the flutter framework generally consists of a combination of small and larger widgets interacting in conjuncture to build the application. All of its layers are integral to its design and functioning. As simple as building an application in flutter is, it is built with equally complex components at its core.